

Programs on boolean literal :

```
void main()  
{  
    boolean isValid = true;  
    boolean isEmpty = false;  
}
```

```
IO.println(isValid);  
IO.println(isEmpty);  
}
```

```
void main()  
{  
    boolean c = 0; //error  
    boolean d = 1; //error  
    IO.println(c);  
    IO.println(d);  
}
```

```
void main()  
{  
    boolean x = "true"; //error  
    boolean y = "false"; //error  
    IO.println(x);  
    IO.println(y);  
}
```

Working with static import :

* This facility importing the static members is available in java from JDK 1.5V
* Using this facility we can directly use static methods without class name.

```
import static java.lang.IO.*;  
  
void main()  
{  
    println("Welcome to static import");  
    String name = readln("Enter your Name :");  
    println("Your Name is :"+name);  
}
```

Signed Integer : Can accept positive and Negative both types of values like byte, short, int, long and so on

Unsigned Integer : Can accept only positive value including 0

Character Literal :

* It is also known as Char literal.

* In Character literal we have only **one data type** i.e. **char data type** which accepts 16 bits. [Supports UNICODE so takes 16 bits of memory]

* We can represent Character Literal in following ways :

a) Single **Character OR Digit OR Special symbol** enclosed in single quotes.

Example :

```
char c1 = 'A';  
char c2 = '5';  
char c3 = '@';
```

```
char ch = 'AB'; //Invalid  
char c1 = ''; //Invalid [Empty character]
```

```
String s = ""; //Valid [Empty String is allowed]
```

b) In older languages like C and C++, which supports **ASCII** value format and range is 0 - 255, On the other hand Java supports UNICODE format where the range is 0 - 65535. [Unsigned integer]

Example :

```
char c1 = 65535; //Valid  
char c2 = 65536; //Invalid
```

c) We can assign character literal to int data type to know the unicode value of that particular character.

Example:

```
int x = 'A';  
IO.println(x); //65
```

d) We can also represent Character literal in 4 digit hexadecimal Unicode format, where the format is :

'\uXXXX'

Here \u represent : Unicode
XXXX represents exactly 4 digits

Range is :
Minimum Range : '\u0000'
Maximum Range : '\uffff'

Example :

97 is a decimal number, If we want to represent in 4 digit hexadecimal format then we need to convert this 97 into hexadecimal and the format is : '\uXXXX'

$$\begin{matrix} (97) & = & (?) \\ 10 & & 16 \end{matrix}$$

16	97	1
16	6	6
	0	

$$\begin{matrix} (97) & = & (061) \\ 10 & & 16 \end{matrix}$$

061 is not a valid hexadecimal format because the format is :

'\uXXXX' but 061 is of 3 digits

```
char ch = '\u0061';  
IO.println(ch); //a
```

e) All the escape sequences like (\n, \t) we can represent using character literal.

Example :

```
char ch = '\n';  
IO.println(ch);
```

//Programs :

```
void main()  
{  
    char ch1 = 'a';  
    IO.println("ch1 value is :"+ch1);  
  
    char ch2 = '5';  
    IO.println("ch2 value is :"+ch2);  
  
    char ch3 = '$';  
    IO.println("ch3 value is :"+ch3);  
}
```

```
void main()  
{  
    int ch = 'A';  
    IO.println("ch value is :"+ch);  
  
    char ch1 = 'A';  
    IO.println("ch1 value is :"+ch1);  
  
    char ch2 = 97;  
    IO.println("ch2 value is :"+ch2);  
}
```

//The UNICODE value for ? character is 63

```
void main()  
{  
    char ch1 = 63;  
    IO.println("ch1 value is :"+ch1);  
  
    char ch2 = 64;  
    IO.println("ch2 value is :"+ch2);  
  
    char ch3 = 65;  
    IO.println("ch3 value is :"+ch3);  
  
    char ch4 = 01;  
    IO.println("ch4 value is :"+ch4);  
}
```

//The UNICODE value for ? character is 63

```
void main()  
{  
    String hindi = "\u0928\u092E\u0938\u094D\u0924\u0947";  
    IO.println(hindi);  
  
    String telugu = "\u0C28\u0C2E\u0C38\u0C4D\u0C24\u0C47";  
    IO.println("\n"+telugu);  
  
    String marathi = "\u0928\u092E\u0938\u094D\u0915\u093E\u0930";  
    IO.println("\n"+marathi);  
  
    String odia = "\u0B28\u0B2E\u0B38\u0B4D\u0B15\u0B3E\u0B30";  
    IO.println("\n"+odia);  
  
    String urdu = "\u0633\u0644\u0627\u0645";  
    IO.println("\n"+urdu);  
}
```

Console X
terminated Test4

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Output from Eclipse IDE

```
void main()  
{  
    int x = 'A';  
    int y = 'B';  
  
    IO.println(x+y); //131  
  
    IO.println('a' + 'a'); //97 + 97 = 194 [Numeric Promotion]  
  
    IO.println("a"+"a"); //aa  
}
```

//Range of UNICODE Value (0 - 65535) OR '\uffff'

```
void main()  
{  
    char ch1 = 65535;  
    IO.println("ch value is :"+ch1);  
  
    char ch2 = 65536;  
    IO.println("ch value is :"+ch2);  
}
```

//WAP in java to describe unicode representation of char in hexadecimal format

```
void main()  
{  
    int ch1 = "\u0000";  
    IO.println(ch1);  
  
    int ch2 = "\uffff";  
    IO.println(ch2);  
  
    char ch3 = "\u0041";  
    IO.println(ch3);  
  
    char ch4 = "\u0061";  
    IO.println(ch4);  
}
```